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THE HISTORY OF

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XANADU

Ted Nelson's fifty-year project for a better hypertext than the one we got

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1960: THE FILE STRUCTURE FOR THE CHANGING

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Ted Nelson -- who coined the words hypertext and hypermedia -- began Project Xanadu in 1960 as a graduate student, imagining a writing system for documents that change: every draft kept, every quotation linked to its source, every connection visible from both ends. His 1965 ACM paper, "A File Structure for the Complex, the Changing and the Indeterminate" [1], introduced the "evolutionary list file" and zippered lists -- the first published design for what he would call hypertext.

Xanadu's requirements hardened into a famous list: documents are permanent and versioned; links are bidirectional and never break; any document can transclude -- include by reference, not by copy -- any part of any other, with visible provenance; and authors are paid automatically through micropayment when their words are transcluded. It is copyright, citation, and publishing rebuilt as a single data structure.

COMPUTER LIB / DREAM MACHINES

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In 1974, unable to wait for the industry, Nelson self-published *Computer Lib / Dream Machines* [2]: two books printed back-to-back and upside down relative to each other, in oversized scrapbook layout. *Computer Lib* told lay readers "you can and must understand computers NOW"; *Dream Machines* catalogued the possible futures -- hypertext, hypermedia, "thinkertoys" -- years before personal computers existed to run them. It became the underground bible of the microcomputer generation and the most direct print ancestor of the web's culture of self-publishing.

Implementation attempts cycled through decades: the 1979 "gold" design at Swarthmore, Autodesk's funding of Xanadu Operating Company (1988-1992), later releases of the code as open source, and demonstration systems like XanaduSpace and OpenXanadu. Wired's 1995 epic "The Curse of Xanadu" [3] pronounced it the longest-running vaporware story in computing; Nelson's reply, "Errors in 'The Curse of Xanadu'", is itself a lesson in document-versus-document dispute that Xanadu links would have made navigable.

THE WEB AS THE WRONG HYPERTEXT

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When Tim Berners-Lee's World Wide Web spread in the 1990s (see the Hypertext chapter), it implemented the cheapest tenth of the hypertext idea: one-way links that break, no versioning, no transclusion, no provenance, copy-paste instead of quotation-by-reference. Nelson has spent the decades since as hypertext's loyal opposition, arguing in books (*Literary Machines*, *Possiplex*) and lectures that the web, PDF, and word processors simulate paper instead of surpassing it.

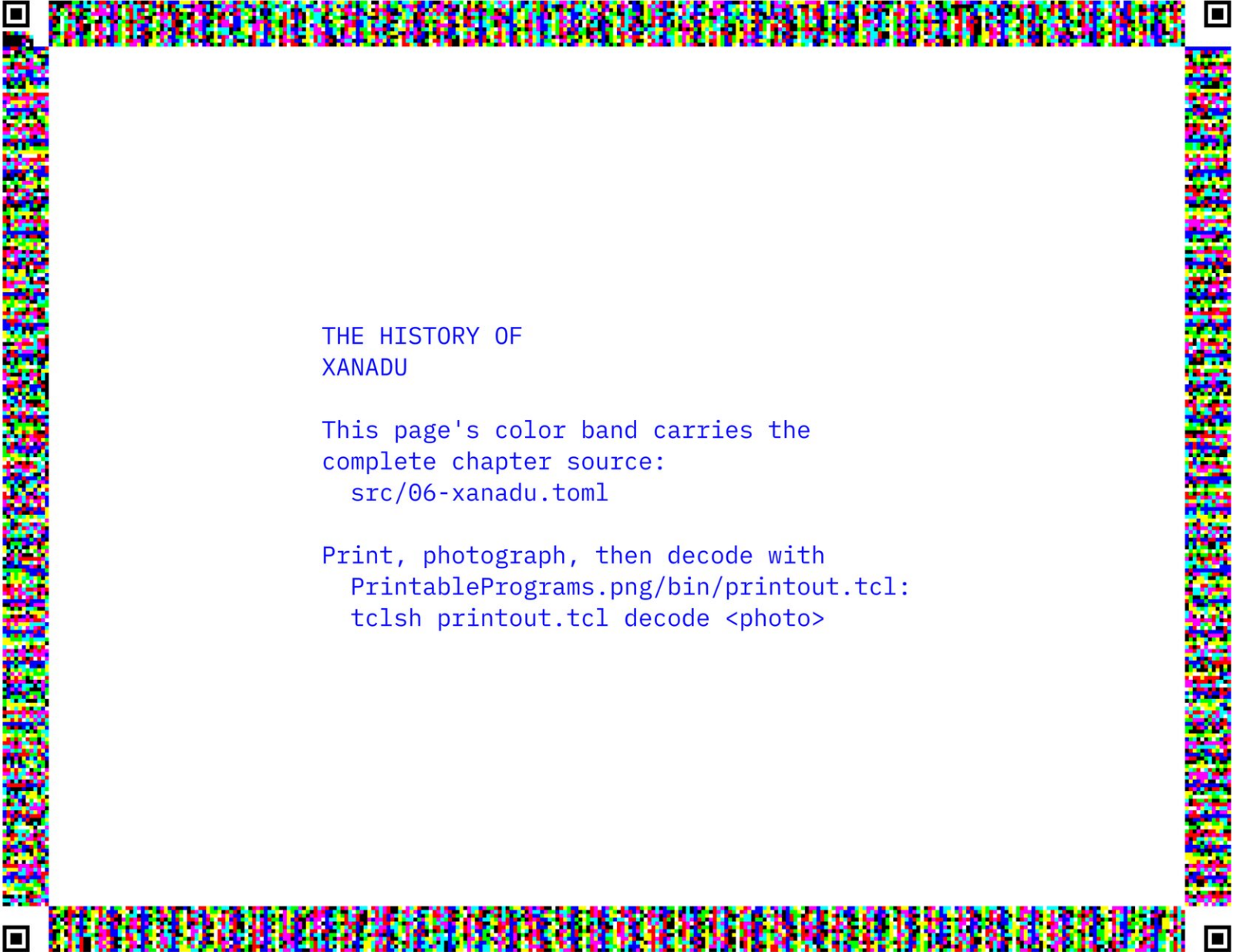
Xanadu's ghost is remarkably productive. Bidirectional links

returned in wikis and in "tools for thought" (Roam, Obsidian); transclusion lives in spreadsheet references, in quote-tweets, in Git submodules; content-addressing and permanent documents are the premise of IPFS and the peer-to-peer hypertexts in this series' Hypertext chapter; micropayment-for-content keeps being reinvented. Project Xanadu persists at xanadu.com [4], still insisting -- correctly -- that the deep hypertext has not yet been built.

SOURCES & CITATIONS

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- [2] Ted Nelson, Computer Lib / Dream Machines (1974), scan on the Internet Archive
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- [4] Project Xanadu, official site
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- [5] Ted Nelson demonstrates OpenXanadu and XanaduSpace, on his YouTube channel (TheTedNelson)
<https://www.youtube.com/@TheTedNelson>
- [6] Vannevar Bush, As We May Think, The Atlantic, July 1945 -- the memex essay Nelson built upon
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